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The Brazilian Pharma & Health Project is an initiative coordinated by the Brazilian Industry Association of Active Pharmaceutical Ingredient (Abiquifi) with the support of the Brazilian Trade and Investment Promotion Agency (ApexBrasil), that aims to expand the participation of companies from Brazilian pharmaceutical and APIs productive chain into the international scenario, seeking to increase exports, technology transfer, investment attraction and the internationalization of the sector.

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Brazil as a Strategic Hub for Clinical Research in Latin America

Consolidated infrastructure, population diversity and regulatory advances position the country as a regional protagonist in clinical trials

Brazil has emerged in recent years as the leading hub for clinical research in Latin America. Currently, the country is responsible for approximately 40% of all clinical studies conducted in the region, twice the volume of countries such as Mexico and Argentina, which rank second and third with around 20% each. This leadership position is not recent; Brazil has built a solid track record in clinical research since the 1980s, with notable contributions to the development of HIV treatments, large-scale clinical trials in diabetes and cardiovascular diseases, and the consistent strengthening of its

research infrastructure. The country benefits from a robust network of experienced research centers and highly qualified investigators, which have positioned it as a strategic partner for global pharmaceutical and biotechnology companies seeking to conduct clinical trials. These insights are highlighted by Fernando Francisco, Executive Manager of the Brazilian Association of Clinical Research Organizations (Abracro), underscoring Brazil's relevance and credibility in the international clinical research ecosystem.

According to Arthur Nigri, Chief Business Officer at Intrials, Brazil distinguishes itself in the global clinical research landscape through a dynamic and contin- ➤





Fernando Francisco,
Executive Manager
of the Brazilian
Association of
Clinical Research
Organizations
(Abracro)

uously evolving ecosystem – characterized by solid scientific expertise, technological advancement, and an increasing emphasis on innovation. He emphasizes that the country has successfully expanded its research infrastructure beyond major metropolitan areas, enabling access to a broader and more demographically diverse patient population. In addition, Nigri highlights key competitive advantages such as cost-effectiveness, ethnic diversity, expedited patient recruitment, and the presence of internationally accredited research centers and investigators. Together, these factors reinforce Brazil's strategic relevance as a destination for multinational clinical trials.

Population diversity is also highlighted by Marina Domenech, Founder and CEO of SAIL for Health, as one of Brazil's greatest strengths in clinical research. According to her, the country offers a rare combination of genetic heterogeneity, distinct metabolic characteristics, and varied social and environmental conditions. This unique mix enhances Brazil's value in the development of personalized therapies that reflect global re-



Marina Domenech,
Founder and
CEO of SAIL
for Health

alities. "The broad variation in metabolic profiles makes clinical trials conducted in Brazil highly representative of diverse populations worldwide, increasing the global relevance and applicability of the results," she emphasizes.

Fernando Francisco reinforces this perspective, noting that Brazil is regarded by many international sponsors as a key player in global clinical research strategies. He points out that the country is not only the largest in Latin America but also benefits from a comprehensive Unified Health System (SUS) and a well-structured network of health services and research infrastructure. "Brazil possesses a robust health infrastructure and is already relatively developed – both essential conditions for conducting high-quality clinical research," he affirms.

Dr. Fernanda Boulos, Medical Director at the Butantan Institute, emphasizes Brazil's leadership in clinical research across Latin America, particularly in the fields of vaccines and biopharmaceuticals. She highlights centers of excellence such as Butantan and Fiocruz, both recognized by the World Health Organization (WHO) among the top global manufacturers. The combination of advanced scientific infrastruc-



Arthur Nigri,
Chief Business
Officer at Intrials

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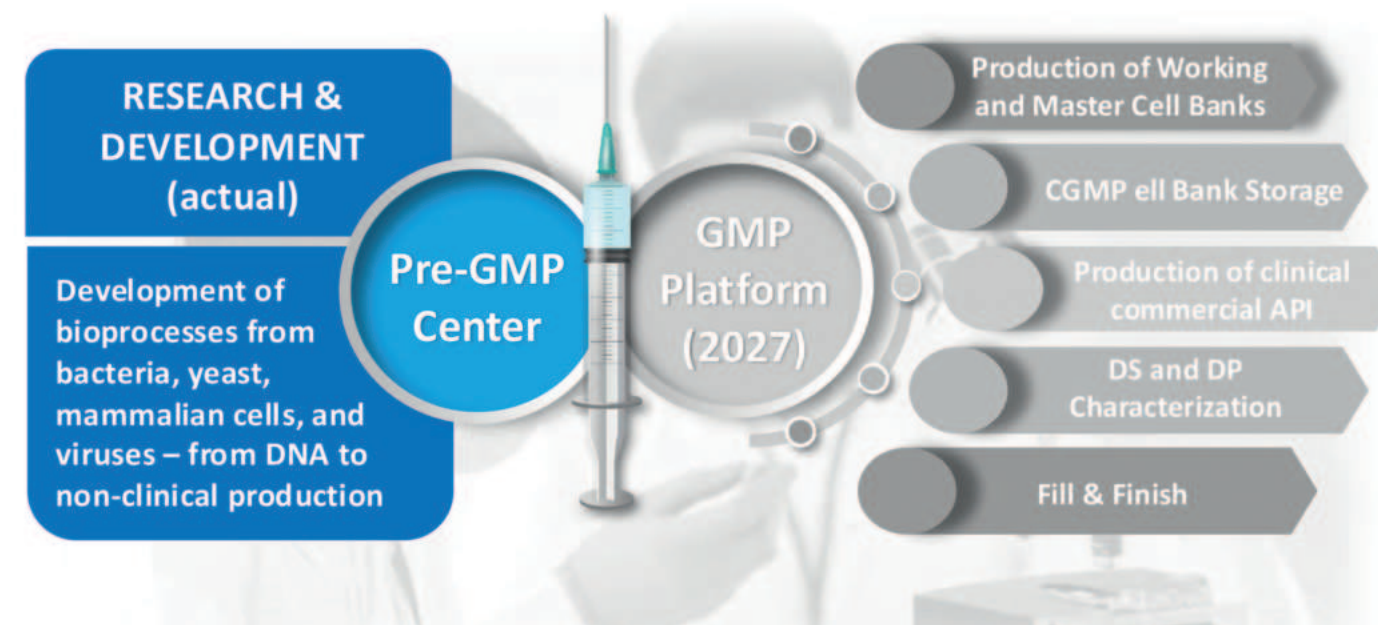


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ture and strategic partnerships with global institutions — such as WHO, NIH, and private sector players — enables Brazil to conduct clinical studies that meet international standards. “We see Brazil as a strategic leader in Latin America, especially in the development of vaccines and biopharmaceuticals,” she states. Fernanda also notes that Brazil’s population diversity and the structure of the Unified Health System (SUS) make the country uniquely suited for large-scale clinical trials, fostering national autonomy, equitable access to healthcare, and inspiration for regional scientific development.

REGULATORY ADVANCES AND INTERNATIONAL ENVIRONMENT

Brazil’s regulatory environment has advanced significantly in recent years. According to Alejandro Arancibia, Medical Director at Amgen Brazil, the enactment of Clinical Research Law No. 14,874 has enhanced legal certainty, improved timelines, and provided new incentives for expanding clinical trials. He also points to ANVISA’s Collegiate Board Resolution (RDC) No. 945/2024, which authorizes the early importation of experimental medicines and allows recognition of assessments by international regulatory agencies — streamlining the approval process when proper storage is ensured until formal authorization is granted.

These developments reinforce Brazil’s position as a trustworthy environment aligned with international regulatory standards. Arthur Nigri adds that such updates are expected to improve approval timelines and regulatory compliance, thereby boosting the country’s competitiveness. Nonetheless, he underscores the importance of continued progress — particularly in reducing bureaucratic delays and fostering public-private partnerships.

Mauricio Zuma, Director of Bio-Manguinhos at Fiocruz, highlights another pillar of Brazil’s strength: the integration of research and production. “Clinical research is a fundamental part of



Mauricio Zuma,
Director of
Bio-Manguinhos
at Fiocruz

Fiocruz’s strategy to provide safe, effective health solutions tailored to the needs of the Unified Health System (SUS),” he states. This integrated approach enables Fiocruz to contribute directly to the development of vaccines and medicines, ensuring scientific excellence, social relevance, and public health impact across Brazil and Latin America.

BRAZIL AS A STRATEGIC MARKET FOR GLOBAL INDUSTRY

Amgen considers Brazil a strategic player in the Latin American clinical research landscape. According to Arancibia, this is due to a combination of factors, including population size, genetic diversity, scientific excellence, and operational capability. With over 200 million inhabitants and a highly mixed population, Brazil generates data that are globally representative. The presence of qualified research centers further strengthens the country’s strategic position. In 2024, Amgen significantly increased its investments in clinical research in Brazil, focusing on oncology, inflammation, and cardiovascular diseases. “Brazil has also demonstrated strong performance in participant diversity. In a cardiovascular study launched in 2023, 42% of Brazilian participants were non-white, ▶



Alejandro Arancibia,
Medical Director at
Amgen Brazil

Dr. Fernanda Boulos, Medical
Director at the
Butantan Institute



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compared to just 17.5% globally,” he notes.

While the private sector sees Brazil as a key market, the public sector faces distinct challenges. Cristiano Gonçalves, Technical Director of Innovation at Butantan, points out that the main barriers to innovation in biologics lie in the pre-clinical phases. “There are structural gaps in producing materials under good manufacturing practices (GMP) and establishing complete value chains that adhere to non-clinical regulatory guidelines and quality standards,” he explains. He emphasizes the importance of programs such as the Local Development and Innovation Program (PDIL), FINEP, and BNDES as critical mechanisms for strengthening Brazil’s innovation and clinical development capabilities.

Marina Domenech adds that startups in Brazil often face regulatory challenges that go beyond bureaucratic hurdles, revealing a systemic disconnect between science, the market, and industrial application. “What many companies fail to grasp is not only the complexity of regulatory procedures, but the absence of a translational approach — turning scientific discoveries into viable, scalable, and sustainable market solutions,” she observes.

BRAZIL’S ROLE AS LEADER AND INTEGRATOR

With the consolidation of ANVISA, the modernization of the legal framework, and its accumulated scientific expertise, Brazil is positioned not only as a leading hub for clinical research but also as a regional integrator. Brazilian leadership can — and should — be exercised collaboratively, leveraging its national capabilities to foster a more cohesive, innovative, and competitive Latin American ecosystem.

Paulo Fernandes, from Azidus, emphasizes that Brazil combines several strategic assets: established scientific capacity, population diversity, and a maturing regulatory framework. The country is home to experienced research centers, highly trained pro-



Cristiano Gonçalves,
Technical Director
of Innovation at
Butantan

fessionals, and a medical education system deeply attuned to the challenges and values of the SUS. He also points out that Brazil’s ethnic representation aligns with the diversity standards required by international regulatory bodies such as the FDA and EMA.

Fernandes believes that the new Clinical Research Law will help Brazil overcome long-standing regulatory bottlenecks and provide a more predictable environment for entrepreneurs, governments, and industry players. “This legal consistency is essential to foster new ventures, launch products, promote public policies, and attract investment,” he concludes.

For Brazil to maintain its upward trajectory in global clinical research, international collaboration and alignment with global demands are key. Mauricio Zuma reinforces this vision: “Fiocruz has strategically worked to strengthen clinical research not only in Brazil, but throughout Latin America — through partnerships with global institutions, regional networks, and multilateral organizations.” This collaborative stance amplifies the impact of research conducted in Brazil and reinforces its role as both a regional and global leader in public health and scientific innovation. •



Paulo Fernandes,
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Innovation in Animal Health:

Biotechnology as a Strategic Driver for National Development

Animal health is gaining prominence in discussions about the future of agribusiness in Brazil. Beyond its direct links to productivity and food quality, it also impacts key areas such as animal welfare, sustainability, and market access. Thus, the need for greater alignment between science, the private market, and public policy is becoming increasingly evident. To better understand how these fronts are interacting, we spoke with Luciano Paulino da Silva, a researcher at Embrapa Genetic Resources and Biotechnology (Cenargen); Rodrigo Secioso, Super-

intendent at the Innovation Brazilian Agency (FINEP); Marcelo de Andrade Mota, Director of the Animal Health Department; and Carlos Goulart, Secretary of Agricultural Defense in the Ministry of Agriculture and Livestock.

Luciano Paulino da Silva highlights the role of green nanotechnology as a sustainability ally in animal health. According to the Cenargen researcher, green synthesis of nanoparticles — particularly silver (AgNPs) and other metals — is being developed from plant extracts from the Cerrado biome in Brazil and agro-industrial waste. These nanoparticles show promising antimicrobial applications in environments such as farms and



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even potential use in cases of animal sepsis, as in vivo trials using murine models indicated no toxicity. “These production routes align with the principles of green chemistry and the Sustainable Development Goals (SDGs), while also reducing environmental impacts and proving economically viable when considering industrial scalability,” he explains.

Paulino also emphasizes the strategic importance of innovation in the country: “The importance of a well-connected innovation ecosystem for animal health lies in its ability to integrate knowledge, resources (human, financial, technological), and a variety of stakeholders, accelerating the development of solutions that balance productivity, sustainability, and animal welfare. Such an ecosystem not only drives technological advances but also strengthens the resilience of the livestock sector and global food security,” he asserts.

Rodrigo Secioso also underscores the importance of partnerships in boosting innovation in animal health. According to him, while the need for coordination varies depending on the type of project and degree of technological disruption, more innovative initiatives generally require stronger collaboration among stakeholders. “Typically, companies have greater capacity to innovate and bring technologies to market, while universities hold the scientific expertise. This partnership creates synergy in promoting an innovation-friendly environment,” he states.

Secioso also highlights the strategic role of startups within this ecosystem, acting as a bridge between science and commercial application. In this context, Embrapa is recognized as a leading player in agricultural technology research, with a strong presence in biotechnology and animal health projects.

Amid this wave of transformation, the Radical Innovation Program (PIR), led by Abiquifi, stands out as a key driver of technological development in animal health. Focused on fostering collaboration among startups, compa-

nies, research institutions, and investors, PIR promotes disruptive initiatives that expand the scientific and technological frontiers of knowledge in Brazil. By supporting advanced research, attracting investment, and facilitating regulatory interaction, the program enhances the development of new products and processes, such as vaccines, innovative pharmaceuticals, and sustainable biotechnologies, contributing to a robust innovation ecosystem. Abiquifi thus plays a strategic role in reinforcing Brazil’s competitiveness in the face of global challenges in animal welfare, sustainability, and food security.

AGRICULTURAL INNOVATIONS: REDEFINING HEALTH AND SUSTAINABILITY

The connection between animal health, sustainability, and welfare is increasingly evident and strategic within the agricultural sector. According to Luciano Paulino da Silva, appropriate management practices, disease prevention, and rational resource use directly impact productivity and environmental conservation. In this context, technologies such as liposomal vaccines, controlled-release drug delivery systems, and 3D bioprinting contribute to reducing chemical usage, enhancing animal welfare, and preserving biodiversity.

Further innovations such as organ-mimetic biofabrication and gamete cryo- ➤

Luciano Paulino da Silva, a researcher at Embrapa Genetic Resources and Biotechnology (Cenargen)





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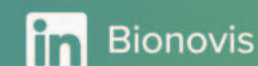
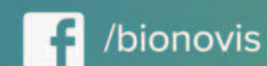
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PHARMACEUTICAL BIOTECHNOLOGY

preservation strengthen herd resilience in the face of climate change and emerging diseases, while sensors and rapid diagnostics enable agile and sustainable responses in sanitary management.

This interdependence between animal health and welfare is also emphasized by Marcelo de Andrade Mota and Carlos Goulart of the Ministry of Agriculture and Livestock (MAPA). “Studies over recent decades have shown a strong association between high health standards and animal welfare indicators, demonstrating that one cannot exist without the other.”

They also note that appropriate infrastructure, efficient management practices, and well-implemented sanitary measures result in healthier animals, reduced reliance on medication, and greater productivity, all of which are key elements for long-term sustainable systems.

Additionally, they highlight the need to adapt regulatory frameworks to keep pace with technological advancements: “The Ministry is continuously developing and updating regulations for veterinary medicines and vaccines. One of the main goals of this effort is to accommodate new technologies, ensuring that product registration processes remain safe and rational.”

Rodrigo Secioso further adds that innovations in animal nutrition, especially those aimed at reducing methane emissions, have gained prominence as environmental mitigation tools, particularly in Brazil, where the cattle herd is significant. He also highlights the importance of water-reduction technologies, waste-to-bioenergy solutions, Active



Rodrigo Secioso, Superintendent at the Innovation Brazilian Agency (FINEP)

Pharmaceutical Ingredients (APIs) development, innovative biological products, and genetic improvements.

According to him, when backed by partnerships among companies, universities, and research centers, these innovations can accelerate the practical application of science, connecting sustainability, animal health, and technology systemically and efficiently.

FINEP plays a catalytic role in driving innovation in the animal health sector. Secioso notes that the institution supports innovation through multiple routes, including investment funds, direct subsidies, and credit mechanisms. “Currently, 37 projects are underway – 17 funded through direct credit to companies and 20 through subsidies – totaling approximately \$265 million allocated to new technology development. In 2024 alone, nine new contracts were signed, with a total of approximately US\$90,8 million in investments. This volume of resources reflects not only

the sector’s potential but also the urgency to modernize it in the light of current health, environmental, and production challenges,” he explains.

Luciano Paulino da Silva concludes by emphasizing that animal health is a strategic pillar of Brazil’s agribusiness, directly tied to global food security and the country’s competitiveness in international markets. He points to innovative initiatives at Embrapa’s Nanobiotechnology.

Laboratory as clear examples of how science and technology can work together toward a more sustainable sector. In addition to nanoparticle research, the lab is advancing disruptive projects such as cultivated meat and plant-based fish analogs, using technologies like 3D bioprinting and plant-derived raw materials.

“These technologies not only reduce the environmental impacts of conventional production but also provide safe, nutritious food alternatives aligned with global consumption and sustainability trends,” he concludes. ●

In addition to nanoparticle research, the lab is advancing disruptive projects such

AS CULTIVATED MEAT AND PLANT-BASED FISH ANALOGS,

using technologies like 3D bioprinting and plant-derived raw materials

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Artificial Intelligence and the New Horizon of Drug Discovery

Recent advances demonstrate the potential of AI to accelerate critical steps in pharmaceutical R&D, from virtual screening to therapeutic personalization

Amid the technological revolution that is transforming different sectors, Artificial Intelligence (AI) has been assuming a leading role in the development of medicines, one of the most complex and fundamental areas for human health. If previously the research and development (R&D) process for new drugs was long, expensive, and often guided by trial and error, today the reality is beginning to change with sophisticated algorithms and advanced predictive models.

At the forefront of this transformation are companies like Ten63 Therapeutics and Servier Brazil, in addition to academic insti-

tutions such as Federal University of Goiás (UFG), through the Center of Excellence in Artificial Intelligence (CEIA). These organizations demonstrate how AI is reshaping every step of the drug development chain, from discovering promising molecules to conducting more accurate and efficient clinical trials.

REVOLUTIONIZING THE PHARMACEUTICAL INNOVATION JOURNEY

Ten63, a North American startup specializing in applying AI to drug discovery, has been using the technology since the beginning of its journey. According to Marcel Frenkel, CEO and Brazilian co-founder of the com-



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Marcel Frenkel,
CEO and co-
founder of the Ten63
Therapeutics

pany, the focus is on the generation and optimization of new molecules.

This approach allows us to explore a virtually infinite chemical space. AI also accelerates calculations based on quantum chemistry, increasing accuracy in predicting molecular interactions and the pharmacokinetic and pharmacodynamic properties of compounds. But Frenkel highlights that human oversight remains essential: "AI is not foolproof: without expert supervision, it can generate misleading predictions. Therefore, we must keep scientists at the center of the decision-making process, using AI as a support and acceleration tool, not as a substitute for human expertise."

ACCELERATE CLINICAL TRIALS AND SHORTEN TIME TO PATIENT

Also focusing on innovation, Servier, an international pharmaceutical group, has been investing in the integration of AI into its R&D processes. To Sandro Albuquerque, Director of Patient Access & Public Affairs at the company in Brazil, technology is already a central part of the strategy: "AI has become an important vector for the innovation process, contributing both to the acceleration of the development of new molecules and to the optimization of both in the design and the analysis of re-



Sandro Albuquerque,
Director of Patient
Access & Public Affairs
Servier Brazil

sults of clinical trials." With millions of potential molecules to be evaluated, AI is fundamental: "Today, we have around 111 million chemical substances cataloged in databases such as PubChem. A small fraction, when analyzed, by combinatorial analysis in the design of molecules. They are about 10 thousand decillion potential bioactive molecules. And this is where AI stands out, due to its ability in volume, speed, and quality of data processing and analysis. (...) We are just in the early days of this new dawn of AI."

APPLIED RESEARCH AT UNIVERSITY

In Brazil, the Federal University of Goiás (UFG), through the Center of Excellence in Artificial Intelligence (CEIA), has demonstrated the positive impact of AI on drug development. According to the teacher Arlindo Galvão, the technology is applied on several fronts:

"AI has been used to discover new bioactive compounds (hits) and optimize prototypes, accelerating and increasing the effectiveness and efficiency of processes. Furthermore, the application of predictive models to evaluate the toxicity of drugs and cosmetics is a prominent area, allowing the anticipation of adverse effects and contributing to the safety of developed products." ➤



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CEIA works with different computational strategies, which include virtual screening, molecular modeling, and simulations.

According to Galvão, the application of AI encompasses steps such as virtual screening of large libraries of molecules, optimization of hits in leads with molecular modeling and computational simulations, in addition to the development of QSAR/QSPR models to predict pharmacokinetic properties, efficacy and toxicity. AI is also beginning to be incorporated into clinical phases, enabling advances in the personalization of treatments.

CROSS-INDUSTRY COLLABORATION DRIVES AI'S IMPACT

A common point among experts is the importance of collaboration for the potential of AI to be realized. To Marcel Frenkel, the synergy between startups, pharmaceutical companies, academia and big tech is essential to transform the disruptive capacity of AI into real solutions. Sandro Albuquerque complements this vision, comparing the current moment to the beginning of the Human Genome Project, like an exponential learning curve. Already Arlindo Galvão reinforces the importance of the triple helix (university, companies and government): "Innovative solutions generated in academia only become sustainable and available to society in new businesses with startups and large industries. This synergy, with the good use of AI, is fundamental to face complex challenges and accelerate drug discovery."

Concrete cases already show the progress of this integration. Servier, for example, maintains partnerships with Google Cloud, Owkin and Atiya, applying AI to models such as digital twins — virtual simulations capable of predicting the impact of medicines before clinical trials. On another front, the company collaborates with startup Elfie for engaging chronic patients with digital coaching technologies.

In the Federal University of Goiás (UFG),



Arlindo Galvão, teacher of the Federal University of Goiás (UFG)

one project stands out: COVID-IA - which accelerated the discovery of drugs against COVID-19 in partnership with institutions from Brazil, South Africa, and Russia. The center also works on neglected diseases such as Malaria, Tuberculosis, Leishmaniasis, and Schistosomiasis, in addition to initiatives to reposition drugs for emerging viruses.

On Ten63, innovation happens in continuous cycles of "predict → test → learn," where data from biochemical assays and simulations are continually integrated to refine AI models, increasing the chances of success and reducing costs.

More than a tool, Artificial Intelligence is becoming a catalyst for a new paradigm in the pharmaceutical sector. By combining unprecedented speed, precision, and analytical power, it paves the way for more personalized, effective, and affordable therapies. There are still ethical, regulatory and technical challenges on the horizon, but the experiences of Ten63, Servier and CEIA/UFG show that it is now possible to envision a future in which drug discovery will no longer be limited by human processing capacity, but expanded by algorithmic intelligence — with the irreplaceable sensitivity and judgment of scientists.●

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An Innovation and entrepreneurship Environment

A technology park of the University of São Paulo and Ribeirão's Preto City hall, one of the most important scientific centers in the country, with the support of several state and federal agencies. It integrates research centers, hard science startups and large companies through the exchange of knowledge and technologies.

Differential Characteristics

Most of the companies installed here were born in the laboratories of the University of São Paulo and that's why they are from Hard Science, intensive in knowledge, with a strong scientific basis. Startups that develop technologies that can transform the industry.

There are tax incentives for companies that are installed here, regarding the exemption from IPTU. We are accredited with CATI of the Ministry of Science, Technology and Innovation, what allows our Technology Center and Startups to receive incentive resources for Computer Science Law.

Within the 70 startups being incubated at SUPERA Park, 13 of them are in the biotech industry. The Incubator program provides these startups with access to top-notch resources and support to help them succeed, including mentorship from experienced professionals and state-of-the-art facilities.

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- We recently launched Container Park, a modern 3,000 m² business center with modular offices, a cafeteria, and shared spaces that encourage collaboration and networking.

Now, we're moving forward with our newest initiative: Health to Business — a center focused on accelerating Agrotech and Biotech startups, reinforcing SUPERA Park's role as a leading hub for innovation and entrepreneurship in Brazil.



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The potential of biodiversity

and international agreements: Brazil's path to global protagonism

Brazil is one of the most biodiverse countries in the world, and is working to transform this natural heritage into innovation with global impact

In recent years, Brazil has taken significant steps toward becoming a global leader in the sustainable use of biodiversity. Advances in regulation, entrepreneurial initiatives, and the consolidation of a robust scientific infrastructure reveal that the country is building the foundation of a thriving ecosystem — capable of transforming its unpar-

alleled biological wealth into innovative, high-impact global solutions.

Recognized as one of the world's most megadiverse nations, Brazil holds not only vast biological resources but also immense cultural richness. This biodiversity represents more than just an ecological treasure, it is a strategic asset for scientific advancement, economic development, and ▶



social transformation. Although challenges remain, a growing number of initiatives are forging a new path by aligning science, business, and socio-environmental responsibility.

In this context, international agreements such as the Nagoya Protocol gain prominence, as they establish a more equitable and sustainable framework for accessing genetic resources and traditional knowledge. According to Peter Andersen, CEO of the Centroflora Group, Brazil must foster a more conducive environment for startups and science-based enterprises working with biodiversity assets.

“Brazil holds 25% of the planet’s biodiversity, yet we still fall short in transforming this into a strategic asset. We need to stimulate solutions involving all stakeholders so that real change can occur,” states Andersen.

This view resonates with the experience of Nintx - Next Innovative Therapeutics, a Brazilian startup focused on identifying new bioactive compounds for pharmaceutical and nutraceutical applications. While acknowledging the complexities of the Brazilian innovation ecosystem, founders Stephani Saverio, Cristiano Guimarães, and Miller Nunes de Freitas are confident in their ability to catalyze transformation. Nintx respects Brazil’s traditional knowledge and genetic resources and believes that the best way to generate, capture and share value from them is through cutting-edge science, developing projects with global attractiveness. In their view, Brazil’s Biodiversity Law of 2015 marked an important step toward organizing the sector. By supporting the continued evolution of the regulatory framework, the country can create a virtuous cycle of investment and benefit sharing with society, enabling more globally competitive projects.

“Regardless of legal mandates, Nintx is committed to ensuring that every new bioactive developed for pharmaceutical or nutraceutical use shares benefits with both the environment and society,” affirms Stephani Saverio.



Peter Andersen,
CEO of the
Centroflora
Group

From the perspective of ecosystem-building organizations, Isabela Allende, Operations and Partnerships Manager at Biominas Brasil, underscores the persistent disconnect between existing legislation and the practical realities faced by entrepreneurs working with biodiversity in Brazil. “We see two main challenges: the first relates to knowledge — both in product development and business modeling. The second is financial viability — many promising products fail due to a lack ▶



Cristiano Guimarães,
Nintx - Next
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of funding, and struggle to prove scalability and production feasibility,” she explains.

To address this gap, Biominas has created both virtual and physical environments to foster connections among projects and start-ups leveraging biodiversity as a strategic asset. One notable initiative is the InovaFittoBrasil Summit, which brings together national and international stakeholders to promote cooperation in research and development.

Despite notable progress, regulatory hurdles persist. Norberto Prestes, Executive President of Abiquifi, acknowledges the country’s scientific capabilities and infrastructure but emphasizes that legal and regulatory adjustments are still needed to attract pharmaceutical industry investments. “We’ve made important advances, but there are still barriers around the Biodiversity Access Law that must be addressed. CNPEM (Brazilian Center for Research in Energy and Materials) is an example of world-class infrastructure — its Sirius electron accelerator is a unique tool that places Brazil at the forefront of R&D globally,” he highlights.

SCIENCE AND SCIENTIFIC INFRASTRUCTURE IN FAVOR OF BIODIVERSITY AND SUSTAINABLE BIOECONOMY

Daniela Trivella (Head of Drug Discovery, LN-Bio-CNPEM), Maria Augusta Arruda (Director, LNBio-CNPEM), and Antônio José Roque da Silva (General Director, CNPEM) also underscore Brazil’s unique position to lead in the sustainable use of biodiversity. The country’s combination of biodiversity, scientific infrastructure, and human capital is unmatched. However, bottlenecks in investment and project scale remain major obstacles.

“Benefit sharing must be seen not just as a legal requirement but as a core component of the scientific process. Improved traceability and mapping of contributions can help refine these mechanisms,” they argue.

Norberto Prestes,
Executive
President of
Abiquifi



LEGAL SECURITY AND TRUST BUILDING

Anita Pissolito, a legal specialist at Nascimento Mourão, emphasizes that while Brazil’s legal framework is based on sound principles, practical tools to support implementation are still lacking. “The absence of regulation of Article 10 of the Nagoya Protocol creates uncertainty, particularly when genetic resources cross national borders. Additionally, each country’s sovereignty over its genetic resources results in a patchwork of national laws, requiring users to navigate varying legislation across jurisdictions,” she explains.

She advocates for a more collaborative model involving traditional communities, research institutions, the private sector, and government. She also stresses the importance of Brazil taking a more assertive role in shaping the international biodiversity agenda.

From an industry standpoint, Giuliano Rodrigo Barissa, Director of New Business and Innovation at Biolab Farmacêutica, offers a pragmatic view of the opportunities and barriers in working with natural assets. “R&D in pharmaceuticals is a time-consuming and costly endeavor, demanding rig-

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Antônio José Roque da Silva, General Director, CNPEM

orous studies and scientific validation. Legal certainty and clear authorization pathways are critical to enabling these investments — especially for national pharmaceutical companies,” he notes.

He also stresses the importance of maintaining an ongoing dialogue with communities and regulators to ensure that innovation is pursued with responsibility and legitimacy.

BIODIVERSITY AND INNOVATION: THE FOUNDATIONS OF BRAZILIAN PROTAGONISM

From these different perspectives, it is clear that Brazil brings together all the necessary elements to transform its biodiversity into a vector of sustainable development: scientific knowledge, entrepreneurial capacity, cultural diversity and biological wealth.

Brazil brings together all the necessary elements to transform its biodiversity into a vector of sustainable development:

SCIENTIFIC KNOWLEDGE, ENTREPRENEURIAL CAPACITY, CULTURAL DIVERSITY AND BIOLOGICAL WEALTH



Giuliano Rodrigo Barissa, Director of New Business and Innovation at Biolab Farmacêutica

The foundations are already being built, with regulatory advances, investments in research infrastructure and innovative initiatives that respect socio-biodiversity.

The success of this transformation now depends on a continuous effort to strengthen the regulatory environment, encourage collaboration between different actors and support the development of solutions that unite science, business and socio-environmental responsibility. With the right alliances and a strategic look at innovation, Brazil has all the conditions to become a global protagonist in the sustainable use of biodiversity. •

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Radical Innovation in Healthcare:

Building a Resilient Ecosystem for the Future

In recent years, Brazil has faced a strategic dilemma: how to transform its scientific capabilities into innovative solutions that positively impact public health and position the country as an attractive hub for global investments.

With this challenge in mind, Abiquifi launched the Radical Innovation Program (PIR) in 2022, aiming to structure a health innovation ecosystem capable of generating pat-

ents, attracting international capital, and retaining talent within the country. The proposal is clear: to create a fertile environment for radical innovation — one that breaks established patterns and offers unprecedented solutions to complex challenges. The PIR brings together various stakeholders — government, industry, universities, and startups — in a coordinated effort to change the innovation landscape in the coming years.

SINCE THE INCEPTION OF THE PROGRAM, SEVERAL IMPORTANT RESULTS CAN BE HIGHLIGHTED:

- The creation of a selection process at ANVISA to evaluate innovative technologies still under development in Brazil, with a view to future registration;
- Launch of a public-private fund focused on health deeptechs;
- Development of a program to prepare Brazilian startups for fundraising in the United States, culminating in the first participation of a national group at the JP Morgan event;
- Training of 12 research centers to expand the discovery and development of new molecules;
- Formulation of a regulatory framework for CDMOs (Contract Development and Manufacturing Organizations), enabling the expansion of technological offerings in the country. ▶

BUT WHAT DOES IT MEAN, IN PRACTICE, TO FOSTER RADICAL INNOVATION IN HEALTH IN BRAZIL TODAY? AND HOW ARE DIFFERENT ECOSYSTEM STAKEHOLDERS CONTRIBUTING TO MAKE THIS VISION A CONCRETE REALITY?

“Mirscience Tx was founded by two PhD candidates who were researching RNA therapies before gaining global recognition. Our focus is on treating diseases that affect longevity, not just living longer, but living better. More than 32 million people die every year from conditions that compromise autonomy and generate billions in healthcare costs. We have developed a technology to keep life in motion. Our molecule, MT-29, increases muscle strength and mass in various models of muscular diseases. It represents a concrete hope for millions of people affected by muscular disorders. With it, we aim to transform treatment approaches and alleviate the burden on healthcare systems. Investing in longevity means investing in quality of life. We remain committed to shaping the future of medicine through science and innovation.”

Lucas Ariel Fernandes da Rocha
Mirscience

“Wecare is a startup that develops products to prevent and treat lesions caused by chemotherapy and radiotherapy, using ingredients derived from biotechnological processes and biodiversity. These lesions affect around 50% of patients, leading to reduced chemotherapy doses and, consequently, lower chances of treatment success. Developing healthcare solutions required navigating complex regulatory and clinical stages, which were made possible through partnerships with research centers such as Unicamp (University of Campinas) and Albert Einstein Hospital. [...] Network collaboration and access to public funding were crucial to bring our products to life, and they remain essential for advancing new active ingredients. Our upcoming challenges include finalizing cost-effectiveness studies and developing six new plant-based pharmaceutical ingredients.”

Cristiane Bertolami
Wecare

“At Biotimize, we believe that radical innovation in healthcare only becomes real when it finds socioeconomically viable paths to move forward. As a CDMO specialized in biopharmaceuticals, we have been directly contributing to turning intellectual property into real therapeutic solutions. We act as a bridge between research and the market, supporting projects from large pharmaceutical companies to startups and universities as they advance by international regulatory standards. In recent years, we have seen a significant increase in both the volume and ambition of innovation projects in Brazil. Our work has been essential to help these initiatives reach non-clinical and clinical stages with safety, quality, and real potential for application.”

Fernando Barbosa
Biotimize

“It is impossible to talk about radical innovation in healthcare without consistent investment in science, technology, and innovation. The Ministry of Science, Technology, and Innovation in Brazil (MCTI) has reaffirmed its commitment to this objective by positioning health as a strategic priority. Following the release of FNDCT (National Scientific and Technological Development Fund), as determined by President Lula, one of the first public calls launched was under the ‘More Innovation Brazil – Health’ program [...] This mobilization reflects the Ministry’s commitment to overcoming the sector’s historical underfunding and promoting significant investments in innovation for the SUS.”

Leandro Pedron
MCTI

Nintx

Next Innovative Therapeutics

Nintx develops a new generation of medicines leveraging biodiversity, primarily from Brazil, targeting multifactorial diseases, caused by genetic and environmental factors as well as their interactions via the human gut microbiome.

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xGlbionics®

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Team A mix of experienced Founders and high-caliber Scientists

PHARMACEUTICALS

NTX-01	Cardiometabolic – indication undisclosed
NTX-02	Infectious Diseases – indication undisclosed
NTX-03	Cardiometabolic – liver/cardiac regeneration
NTX-04	Gastroenterology – irritable bowel syndrome
NTX-05	Immunology – immuno-oncology
NTX-06	Gastroenterology – inflammatory bowel disease
NTX-07	Neuroscience – Alzheimer’s Disease
NTX-08	Multiple TAs – indication undisclosed

NUTRACEUTICALS

PBTX-01	Newborn gastrointestinal health
PBTX-02	Digestive support for non-celiac gluten sensitivity
PBTX-03	Broad-spectrum digestive support for adults
PBTX-04	Irritable bowel syndrome
PBTX-05	Gut barrier restoration
PBTX-06	Digestive support for NCGS + Gut barrier restoration
PBTX-07	Microbiome gut-brain axis modulation
PBTX-08	Histamine intolerance

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“Driving innovation in healthcare regulation in Brazil — a commitment undertaken by Anvisa — will benefit the Brazilian healthcare sector, especially the Brazilian Unified Health System (SUS), which, through health surveillance, serves approximately 215 million people daily, 100% of the Brazilian population. Surveillance regulates nearly 250 economic activities, ensuring access to and safe consumption of essential health products and services. Considering the rise of new technologies, especially those involving automation and artificial intelligence, advancing quickly in this area is crucial for contributing to national development. [...] We live in a highly technological world, where artifacts with direct impacts on individual and collective health constantly emerge, requiring the integration of a growing number of products, services, and information that empower citizens to make better health choices. In this context, our challenge is to keep pace with this speed.”

Artur Iuri Alves de Sousa
Anvisa

“Finep (Innovation Brazilian Agency) plays a vital role in the national health innovation system. [...] The ‘More Innovation Brazil’ program, implemented by Finep, seeks to support the development of innovative projects in the industrial sector in the following areas: development of active pharmaceutical ingredients (APIs), development of biological products with high impact for the Brazilian Unified Health System (SUS), clinical research for domestic innovation, and priority products defined by the Ministry of Health. [...] So far, the program has led to the contracting of 19 projects, totaling approximately US\$37 million.”

Igor Bueno
Finep

“The Radical Innovation Program led by Abiquifi represents a strategic initiative aligned with Libbs’ mission to contribute to the sustainable development of the pharmaceutical value chain in Brazil. [...] Libbs has made significant investments in global startups that are developing innovative therapies with the potential to transform the patient journey. These investments total nearly USD 20 million and are expected to be allocated to startups over the next three years. Additionally, we are supporting Brazilian startups by scaling up their production chains to enable clinical development of their products.”

Carlos Eduardo Vitor
Libbs

“Biomina Brasil is an open innovation hub in the healthcare sector, promoting entrepreneurship and innovation throughout the development chain of new products and services.

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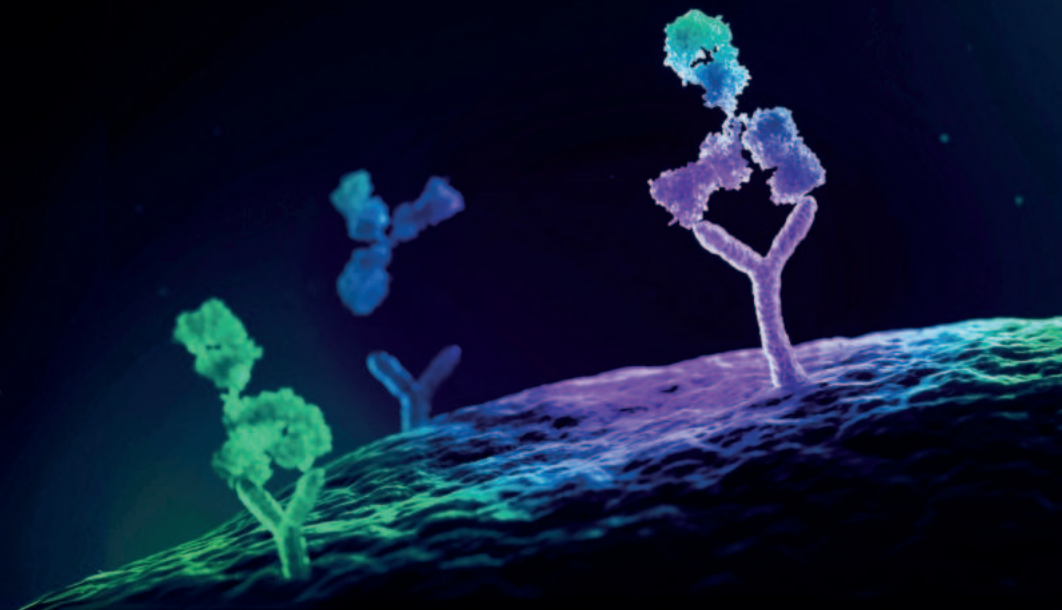


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Population aging and chronic diseases: how Brazil will face this challenge

By **Milton Crenitte** and
Alexandre Kalache



LONGEVITY AS A SOCIAL REVOLUTION: BRAZIL'S PATH TO AN AGING FUTURE

The longevity we witness today in Brazil and worldwide represents a true revolution. If we define a revolution as a sudden, irreversible shift capable of transforming all aspects of a society, then population aging unquestionably fits this definition. The increase in the proportion of older individuals is extraordinary. In 2010, just over 10% of the Brazilian population was aged 60 or over; by 2022, this figure had risen to 15.6%, representing 32,113,490 people. By 2050, projections estimate that number will reach 68 million, more than 30% of the population. We are aging at an unprecedented pace. In just two decades, between 2011 and 2030, Brazil will undergo demographic changes that took countries like France and the United King-

dom over a century to experience.

This is, without question, one of the greatest social achievements of the past hundred years. However, despite the progress and the celebration of aging as a new norm rather than an exception, there is another side to this coin: the rapid rise of non-communicable chronic diseases (NCDs), such as hypertension, diabetes, cardiovascular diseases, cancer, musculoskeletal disorders, and dementias.

According to the World Health Organization, more than 70% of deaths worldwide are due to these conditions. In Brazil, aging is not only premature but often accompanied by poor health. Long before reaching what is traditionally considered "old age," a significant portion of the population already presents with one or more chronic illnesses. This reality was starkly exposed during the

COVID-19 pandemic. The explanation is straightforward: Brazil's population is aging within a context of deep social inequality, which further exacerbates health risks. It is not population aging per se that is problematic, but the conditions under which a large portion of Brazilians age prematurely and with limited support. Addressing this scenario requires planning, political will, and, above all, a cultural shift in how we approach care.

BRAZIL'S PROGRESS: WHAT WE ARE DOING WELL

Despite the significant challenges, it is important to recognize Brazil's achievements. The country is home to the world's largest universal public health system: the SUS (Unified Health System). This represents a true civilizational milestone. SUS not only guarantees free access to health-



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care services but also breaks away from a clientelist model, one based on favors, patronage, or humiliation to receive care. With its creation, access to health became a right for all – a transformative and redemptive shift.

Over the past decades, SUS has incorporated robust primary care and disease prevention policies, including the National Health Policy for the Elderly (PNSPI), expansion of the Family Health Strategy, and vaccination programs. Yet more is needed, including intersectoral strategies. Health is built in daily life – where people live, work, travel, socialize, and love – not just within the four walls of a clinic.

That said, SUS faces real challenges, including underfunding – and at times, even defunding – which jeopardizes its core principles: universality, comprehensiveness, and equity. The solution lies in strengthening and defending SUS, ensuring that interests misaligned with the public good do not undermine it.

INNOVATIONS IN LONGEVITY IN BRAZIL

Brazil offers a promising environment for innovation in longevity, both through technological advances such as gerontechnology and through initiatives aimed at improving elder care and inclusion. These include:

- **Age-friendly cities:** Some municipalities have adopted the WHO's active aging framework to adapt public spaces related to transport, leisure, and social engagement. These changes promote social inclusion, health, safety, and lifelong learning for older adults.

- **New housing and co-living arrangements:** From village-style housing and cohousing to community cen-

ters and other forms of shared living, such initiatives offer business opportunities and create welcoming spaces where all individuals feel valued, respected, and included.

- **Technologies supporting person-centered care:** Gerontechnology is gaining traction with the development of apps, smart devices, AI-based tools, and platforms for telemonitoring, medication management, and preventive health promotion.

- **Interdisciplinary collaboration:** Brazil has promising models such as Oncogeriatrics at institutions like the São Paulo State Cancer Institute, where geriatricians collaborate with oncologists to guide therapeutic decisions, minimize treatment toxicity, and better identify candidates for palliative care.

- **Education and empowerment:** Brazilian universities like the University of São Paulo and the State University of Rio de Janeiro offer courses open to older adults, fostering both learning and intergenerational integration. Lifelong learning is a critical pillar of active aging.

The pharmaceutical industry also plays a key role in innovations targeting chronic diseases that disproportionately affect older adults. Advances in biotechnology, targeted therapies, and extended-release medications have significantly improved the management and quality of life of patients with osteoporosis, cardiovascular conditions, diabetes, and certain cancers. Companies have invested in clinical research, partnerships with centers of excellence, predictive technologies, and early diagnosis and treatment adherence programs, promoting personalized medicine in healthy longevity support.

INEQUALITY, WORKFORCE GAPS, AND THE ROLE OF GERIATRICS

There are many reasons why health-care professionals in Brazil are not adequately prepared for an aging population. The life-course perspective – essential for understanding care at every stage – is often absent from healthcare education. For instance, only 10% of Brazilian medical schools offer a geriatrics course.

How can a future doctor, physiotherapist, or psychologist become passionate about something they never had the chance to study? In 2024, among medical specialties chosen for residency (mostly by young physicians), approximately 11% selected Obstetrics/Gynecology, a similar number chose Pediatrics, and only 0.4% opted for Geriatrics. And yet aging changes everything – from anatomy, physiology, and pathophysiology to drug metabolism, interactions, and even disease presentation.

All future health professionals should be exposed to modules and content that prepare them for population aging. Without this “Educational Revolution,” we will continue training professionals “for the 20th century” – focused on child development and maternal health – to treat older adults with multiple chronic conditions. The consequences? Mistakes and negative attitudes that are often reflected in remarks like “What a day! That’s the tenth elderly patient I’ve seen.”

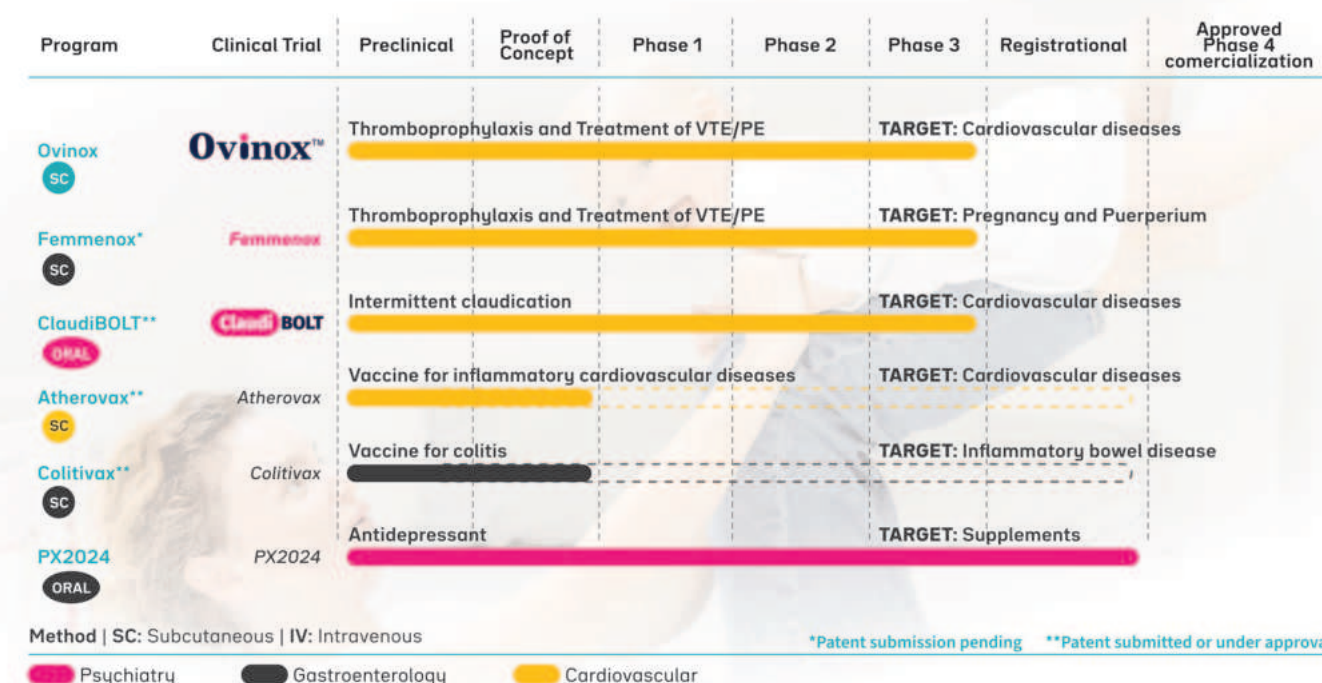
These mistakes can be serious. Iatrogenesis already ranks as the second or third leading cause of death among older adults in Brazil, often hidden by corporatist interests. Lack of training can and does harm patients. ➤

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The Brazilian Society of Geriatrics and Gerontology (SBGG) estimates a shortage of 28,000 geriatricians in the country. Even if we trained 5,000 new specialists over the next decade, demand would still far outpace supply.

OLDER ADULTS ARE NOT "JUST OLDER ADULTS"

Central to geriatrics is the understanding that older adults have unique metabolic and physiological characteristics. The process of senescence, the natural biological aging we all undergo, changes the body's water-to-fat ratio and alters the function of vital organs such as the kidneys, lungs, heart, liver, skin, and brain.

This explains why new symptoms in older adults should always be treated as potential warning signs of medication side effects. Drug absorption, distribution, metabolism, and excretion occur differently from those in younger individuals.

As a result, iatrogenic harm, as previously discussed, becomes more frequent, especially when healthcare professionals fail to consider these distinctions. Worse still, inappropriate interventions may lead to fatalities in older patients without the physician even realizing the harm caused.

THE TRANSFORMATIVE ROLE OF OLDER ADULTS

Aging is not synonymous with decline but with potential. Active, healthy, and autonomous older adults with access to opportunities are invaluable assets to their families, communities, the economy, and the nation as a whole. In a rapidly aging Brazil with a shrinking young workforce, ensuring that older adults have access to health-

care, continuing education, social participation, and security is a key strategy for national sustainability.

We are discussing building a sustainable future, not fueling a generational divide. After all, today's youth aspire to grow old and hope to do so in a country ready and willing to support them.

It is essential to envision a future in which older adults continue to work, create, care, and lead. More importantly, this must not be a privilege for a few but part of a national project. To achieve this, we must confront ageism – prejudice against older people. Ageism perpetuates neglect, dehumanization, and inadequate public policies.

WITHOUT COMBATING AGEISM, THERE IS NO VIABLE FUTURE

Aging is the defining challenge of the 21st century. Every effort must be made to ensure that this process unfolds with quality and dignity. To this end, we must foster a true culture of care – one in which men share caregiving responsibilities; in which aging is valued from an early age; and in which public policies are intersectional, treating health as something built through networks, daily routines, and community engagement.

Brazil has a long road ahead. And it begins with the recognition that to age is to live, and that living well must be a right for all. •

It is essential to envision a future in which older adults continue to work, create, care, and lead.

MORE IMPORTANTLY, THIS MUST NOT BE A PRIVILEGE FOR A FEW BUT PART OF A NATIONAL PROJECT



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Acceleration of innovation and the legacy of COVID-19:

Bio-Manguinhos/Fiocruz as a reference in the global response to future health threats

By **Mauricio Zuma** and
Rosane Cuber

In 2025, it will be five years since the COVID-19 pandemic transformed the world. The social, economic, and health impact of the crisis was profound, but it also brought valuable lessons that drove significant advances in science, technology, and public health. Amid this scenario, Bio-Manguinhos/Fiocruz assumed a central role in the Brazilian response to the pandemic, consolidating itself as a strategic institution for national and international public health.

During the peak of the crisis, Bio-Manguinhos' actions were decisive: the production of vaccines in record time, the manufacture and distribution of diagnostic kits and personal protective equipment guaranteed the country's autonomy and response capacity. The partnership

with AstraZeneca and the consolidation of 100% national production of the vaccine against COVID-19 demonstrated the Institute's technical, scientific and regulatory capacity.

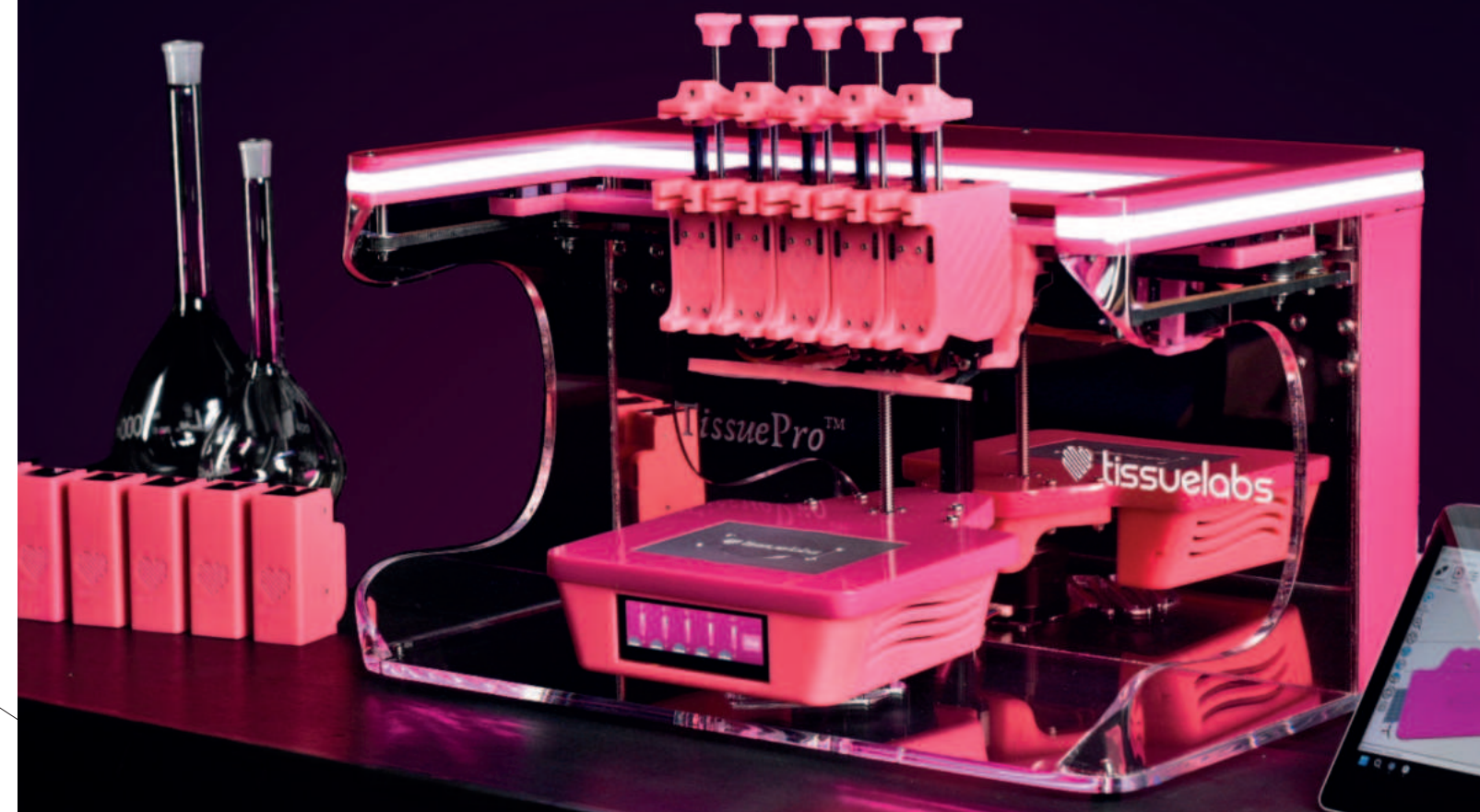
More than an emergency response, the pandemic left a lasting legacy. Since then, Bio-Manguinhos has continually invested in research, development, and innovation, paving the way for an even more agile and efficient response to future pandemics. One of the milestones of this process is adherence to the 100-Day Mission, an international initiative led by the International Pandemic Preparedness Secretariat (IPPS), which sets the objective of developing and making available a safe and effective vaccine within 100 days after the identification of a new pandemic threat.

In this sense, Bio-Manguinhos is part of a global network of strategic centers that are at the forefront of the development of advanced technological platforms, such as messenger RNA (mRNA). In 2021, the institution was selected as an mRNA technology Hub for Latin America and the Caribbean, within the scope of the WHO Technology Transfer Program and the Medicines Patent Pool (MPP). This designation not only recognizes the technical-scientific excellence of Bio-Manguinhos, but also expands its responsibility in building regional capacities and strengthening the collective response to health emergencies.

The application of mRNA technology goes beyond COVID-19. Bio-Manguinhos is already studying its use for vaccine development against ne- ➤

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glected diseases, such as yellow fever, and for therapeutic applications, such as cancer treatment, and other diseases. This innovation is accompanied by another promising advance: the implementation of advanced therapy platforms, such as CAR-T technology. In partnership with the American organization Caring Cross, Bio-Manguinhos is building the first modular unit for the national production of CAR-T cells, which will allow the treatment of leukemias and lymphomas with cutting-edge technology and with a focus on equitable access through the Brazilian Unified Health System (SUS).

These efforts are part of a broader strategy of technological sovereignty and strengthening of the Health Economic-Industrial Complex in Brazil (CEIS). Bio-Manguinhos' work with the network of vaccine manufacturers in developing countries (DCVMN), the WHO, and initiatives such as the Global Vaccine and Immunization Research Forum (GVIRF) has reinforced Global South cooperation, promoting the exchange of knowledge, technology transfer, and the coordination of clinical trials.

The construction of the Santa Cruz Campus (CIBS), which will be the largest vaccine production center in Latin America, is another concrete step in expanding national production capacity. The new unit will accommodate the production of vaccines with mRNA and viral vector technology, making Brazil more prepared to lead rapid and effective responses to future threats.

In addition to the structural milestones, the Institute continues to expand its technological development portfolio. In 2025, two new projects

will gain prominence: the combined measles, rubella, and yellow fever vaccine (MR-YF) and the RNA Network, a knowledge transfer initiative for the development of self-amplifying RNA (saRNA) and conventional mRNA systems, both with support from the Gates Foundation.

Bio-Manguinhos/Fiocruz's commitment to public health goes beyond scientific innovation. It is at the heart of its institutional mission to guarantee equitable access to quality vaccines and treatments, promoting a fairer and more sustainable healthcare system. The COVID-19 pandemic has left deep scars, but it has also catalyzed an unprecedented transformation. By investing in science, strengthening partnerships, and expanding its production capacity, Bio-Manguinhos/Fiocruz is consolidating itself as a global refer-

ence in responding to health emergencies and preparing for the future.

We have a clear mission: to be ready to develop and deliver a vaccine in 100 days, and Bio-Manguinhos is preparing for this challenge. •

Bio-Manguinhos/Fiocruz's commitment to public health goes beyond scientific innovation. It is at the heart of its institutional mission to guarantee equitable access to quality vaccines and treatments,

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